

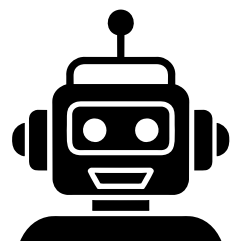
LLM Scoring

Tabular dataset

sex	age	...	credit	label
male	67		1159	1
female	22		5257	0
female	49		3452	1

-  Protected attribute (A)
-  Features (X_j)
-  Label (y)

Prompt Serialization



LLM

Decision Score
 $s(\mathbf{x}) \in [0, 1]$

Phase 1: Baseline



Group Fairness

SP

Statistical Parity

EOD

Equalized Odds

PP

Predictive Parity

Individual Fairness

TI

Theil Index

Phase 2: Direct Unfairness



Mask protected attribute

sex	...	label
UNKNOWN		1
UNKNOWN		0
UNKNOWN		1

LLM Scoring

Decision Score
 $s(\mathbf{x}_{-A}) \in [0, 1]$

Direct Unfairness
score

$$\mathbb{E}_{\mathbf{x}} |s(\mathbf{x}) - s(\mathbf{x}_{-A})|$$

Phase 3: Proxy Risk Analysis

Proxy Capacity

Information association

$$PC_j = I(X_j; A)$$

Proxy Use

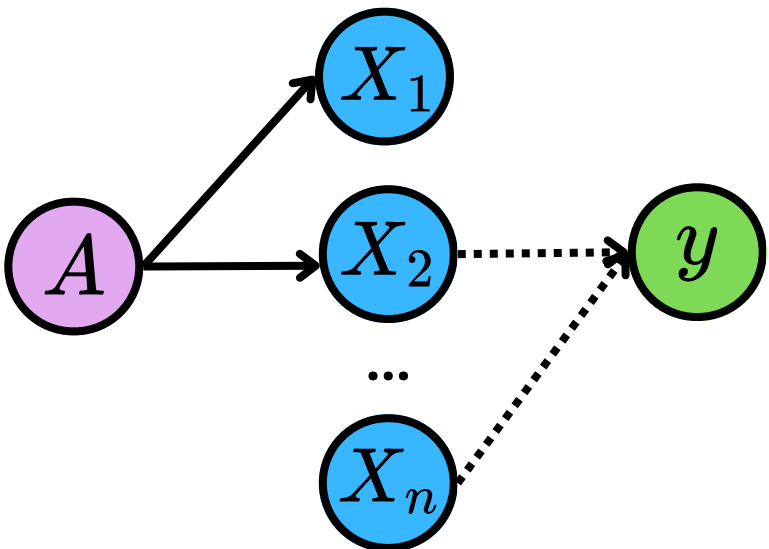
Effect of ablation feature

$$PU_j = \mathbb{E}_{\mathbf{x}} |s(\mathbf{x}) - s(\mathbf{x}_{-j})|$$

Proxy Risk Score

$$PRS_j = PC_j \times PU_j$$

Proxy Path Graph



Phase 4: Mitigation

Proxy Masking

...	...	...	...
...	UNKNOWN		...
...	UNKNOWN		...
...	UNKNOWN		...

 High-risk proxy features

System Prompt Debiasing

You are an impartial evaluator. Do not infer or rely on demographic characteristics ...